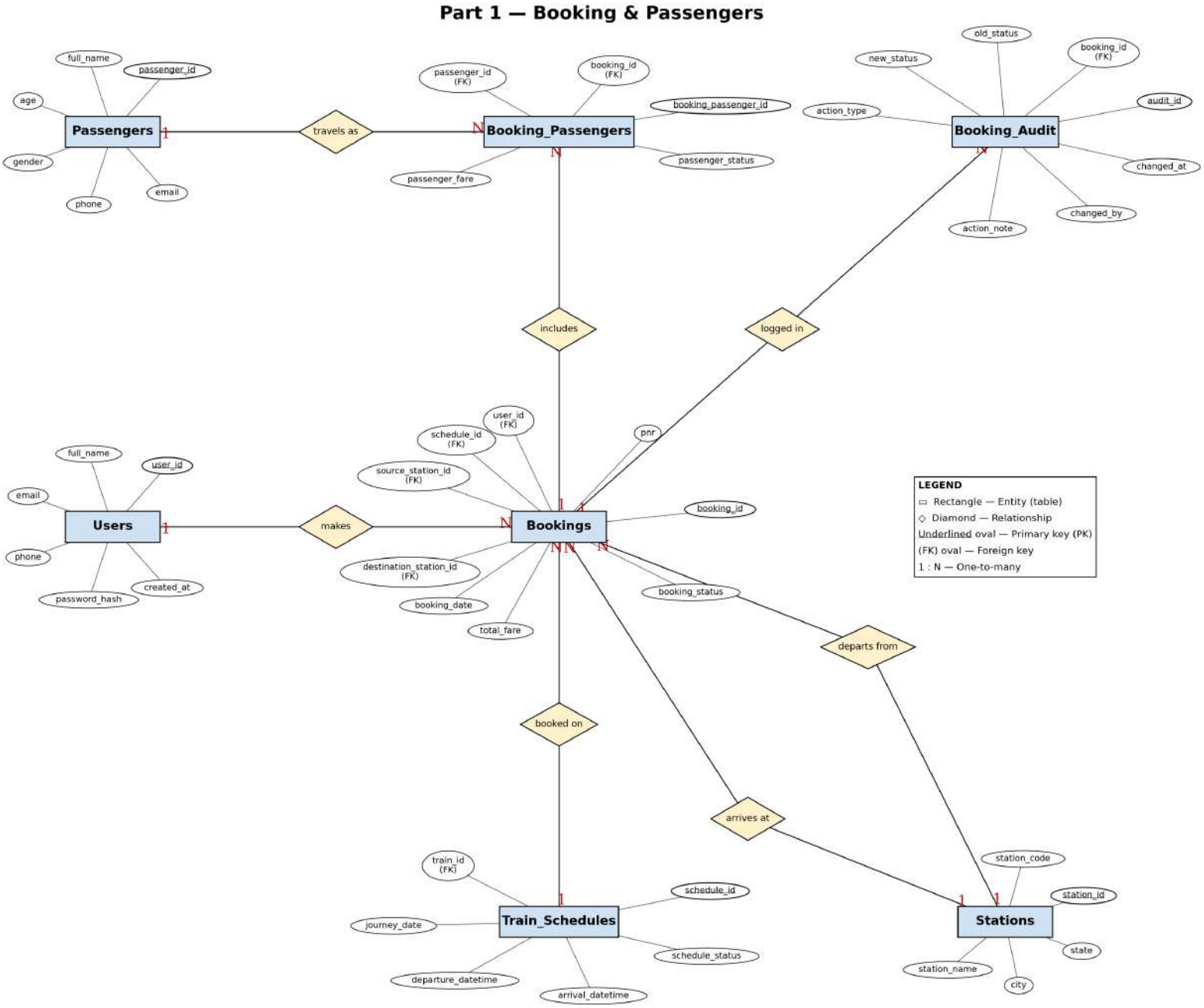
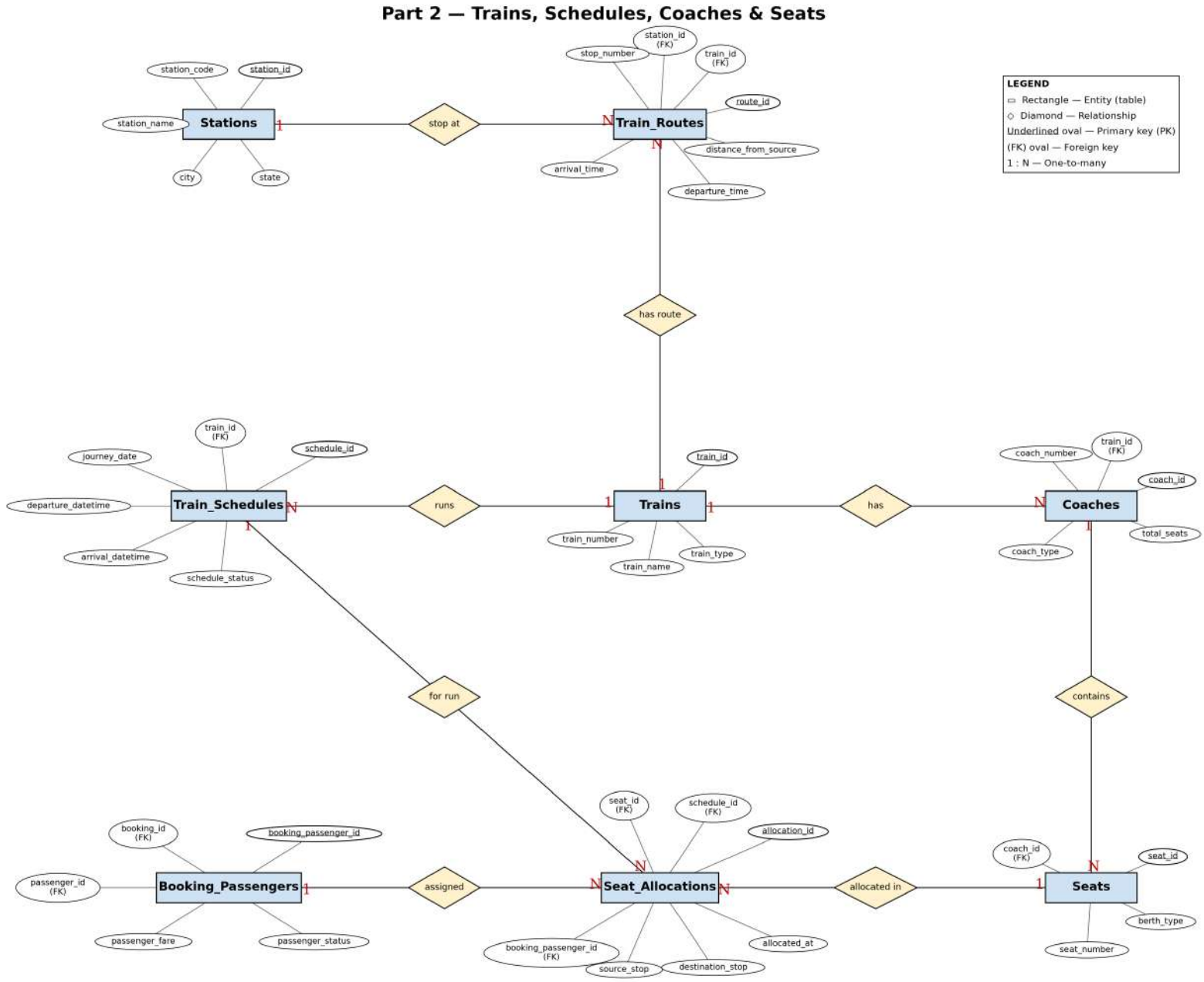
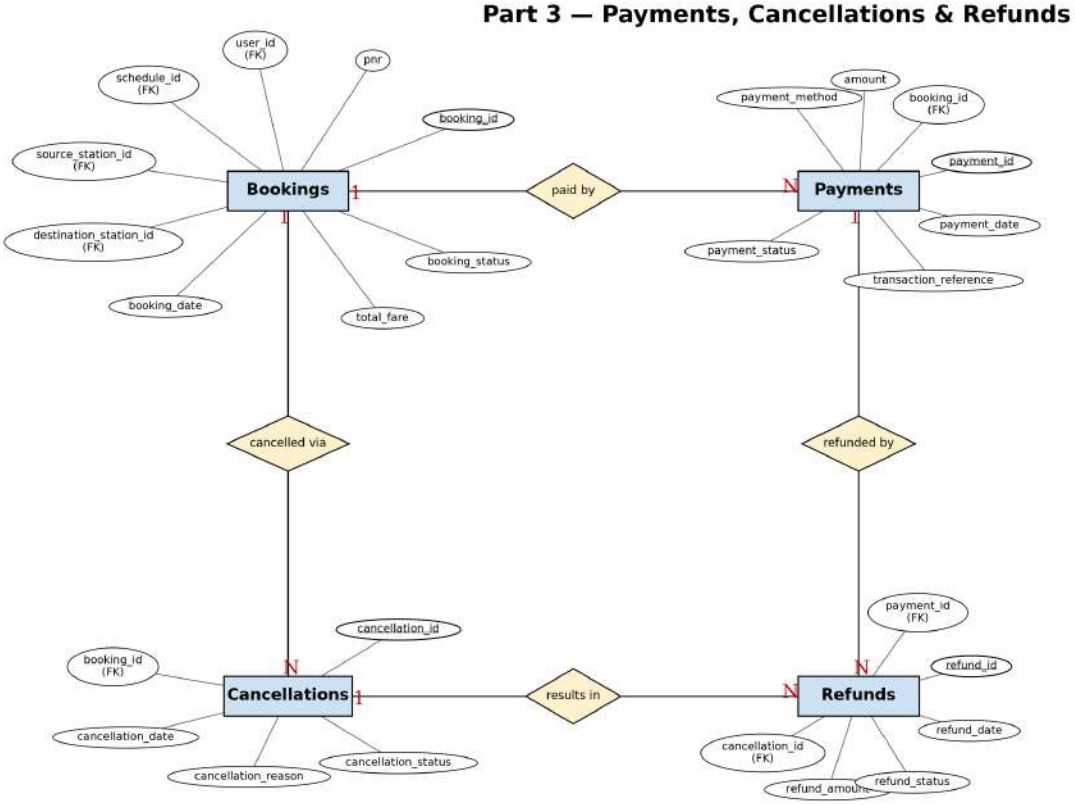

Part 1 — Booking & Passengers
Passengers
full_name
passenger_id
age
gender
phone
email
travels as
Booking_Passengers
passenger_id (FK)
booking_id (FK)
booking_passenger_id
passenger_fare
passenger_status
Booking_Audit
old_status
new_status
booking_id (FK)
audit_id
action_type
changed_at
changed_by
action_note
includes
logged in
Users
full_name
user_id
email
phone
password_hash
created_at
makes
Bookings
user_id (FK)
schedule_id (FK)
source_station_id (FK)
pnr
booking_id
destination_station_id (FK)
booking_date
total_fare
booking_status
departs from
booked on
arrives at
Train_Schedules
train_id (FK)
schedule_id
journey_date
departure_datetime
arrival_datetime
schedule_status
Stations
station_code
station_id
state
station_name
city
LEGEND
Rectangle — Entity (table)
Diamond — Relationship
Underlined oval — Primary key (PK)
(FK) oval — Foreign key
1 : N — One-to-many

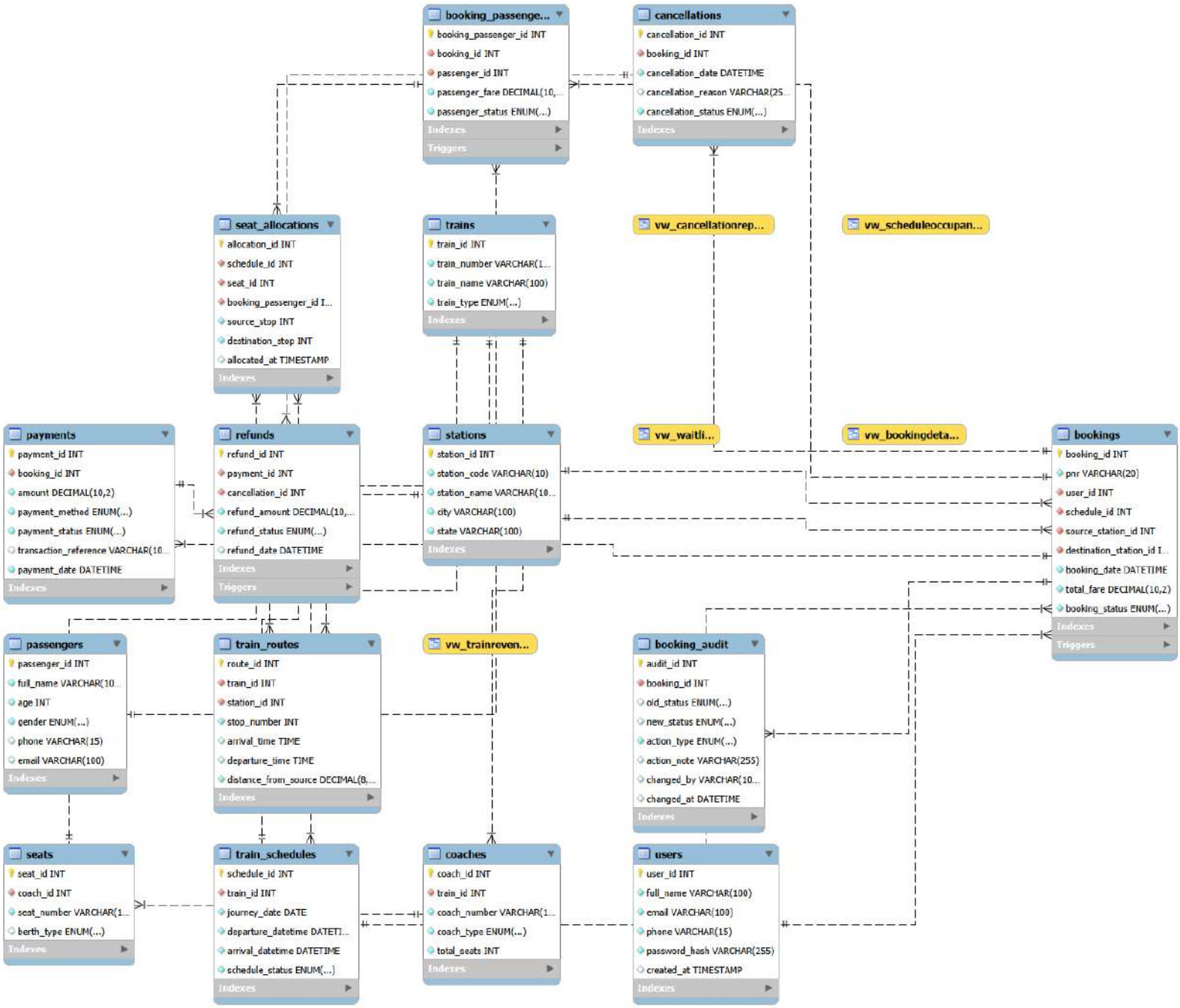

Part 2 — Trains, Schedules, Coaches & Seats
LEGEND
Rectangle — Entity (table)
Diamond — Relationship
Underlined oval — Primary key (PK)
(FK) oval — Foreign key
1 : N — One-to-many
Stations
station_code
station_id
station_name
city
state
stop at
Train_Routes
station_id (FK)
stop_number
train_id (FK)
route_id
distance_from_source
arrival_time
departure_time
has route
Train_Schedules
train_id (FK)
journey_date
schedule_id
departure_datetime
arrival_datetime
schedule_status
runs
Trains
train_id
train_number
train_name
train_type
has
Coaches
coach_number
train_id (FK)
coach_id
total_seats
coach_type
for run
contains
Booking_Passengers
booking_id (FK)
booking_passenger_id
passenger_id (FK)
passenger_fare
passenger_status
assigned
Seat_Allocations
seat_id (FK)
schedule_id (FK)
allocation_id
booking_passenger_id (FK)
source_stop
destination_stop
allocated_at
allocated in
Seats
coach_id (FK)
seat_id
berth_type
seat_number

Part 3 — Payments, Cancellations & Refunds



LEGEND

- ▭ Rectangle — Entity (table)
- ◇ Diamond — Relationship
- Underlined oval — Primary key (PK)
- (FK) oval — Foreign key
- 1 : N — One-to-many


booking_passenge...
booking_passenger_id INT
booking_id INT
passenger_id INT
passenger_fare DECIMAL(10,...
passenger_status ENUM(...)
Indexes
Triggers
cancellations
cancellation_id INT
booking_id INT
cancellation_date DATETIME
cancellation_reason VARCHAR(25...
cancellation_status ENUM(...)
Indexes
seat_allocations
allocation_id INT
schedule_id INT
seat_id INT
booking_passenger_id I...
source_stop INT
destination_stop INT
allocated_at TIMESTAMP
Indexes
trains
train_id INT
train_number VARCHAR(1...
train_name VARCHAR(100)
train_type ENUM(...)
Indexes
vw_cancellationrep...
vw_scheduleoccupan...
payments
payment_id INT
booking_id INT
amount DECIMAL(10,2)
payment_method ENUM(...)
payment_status ENUM(...)
transaction_reference VARCHAR(10...
payment_date DATETIME
Indexes
refunds
refund_id INT
payment_id INT
cancellation_id INT
refund_amount DECIMAL(10,...
refund_status ENUM(...)
refund_date DATETIME
Indexes
Triggers
stations
station_id INT
station_code VARCHAR(10)
station_name VARCHAR(10...
city VARCHAR(100)
state VARCHAR(100)
Indexes
vw_waitli...
vw_bookingdeta...
bookings
booking_id INT
pnr VARCHAR(20)
user_id INT
schedule_id INT
source_station_id INT
destination_station_id I...
booking_date DATETIME
total_fare DECIMAL(10,2)
booking_status ENUM(...)
Indexes
Triggers
passengers
passenger_id INT
full_name VARCHAR(10...
age INT
gender ENUM(...)
phone VARCHAR(15)
email VARCHAR(100)
Indexes
train_routes
route_id INT
train_id INT
station_id INT
stop_number INT
arrival_time TIME
departure_time TIME
distance_from_source DECIMAL(8,...
Indexes
vw_trainreven...
booking_audit
audit_id INT
booking_id INT
old_status ENUM(...)
new_status ENUM(...)
action_type ENUM(...)
action_note VARCHAR(255)
changed_by VARCHAR(10...
changed_at DATETIME
Indexes
seats
seat_id INT
coach_id INT
seat_number VARCHAR(1...
berth_type ENUM(...)
Indexes
train_schedules
schedule_id INT
train_id INT
journey_date DATE
departure_datetime DATETI...
arrival_datetime DATETIME
schedule_status ENUM(...)
Indexes
coaches
coach_id INT
train_id INT
coach_number VARCHAR(1...
coach_type ENUM(...)
total_seats INT
Indexes
users
user_id INT
full_name VARCHAR(100)
email VARCHAR(100)
phone VARCHAR(15)
password_hash VARCHAR(255)
created_at TIMESTAMP
Indexes